

TAMILNADU POWER GRID CORPORATION

Tirunelveli Sub-Station — Bay 7 (132/33 kV Grid Transformer)

POWER TRANSFORMER — ANNUAL ELECTRICAL INSPECTION REPORT

Report No.	TNPGC-ELEC-2026-0712	Date	10-Sep-2026
Asset Tag	TX-BAY7-132KV	Location	Tirunelveli Sub-Station, Bay 7
Equipment Type	Power Transformer 132/33 kV, 50 MVA	Work Order	WO-2026-TX712
Manufacturer	BHEL (Bhopal)	Year of Manufacture	2011
Vector Group	YNd11	Cooling Type	ONAN / ONAF
Inspector	K. Anburaj (Sr. Electrical Engineer)	Emp ID	EMP-40218
Next Inspection Due	Sep-2027	Approval	Chief Engineer R. Balaji

1. Transformer Overview

Transformer TX-BAY7-132KV is a 50 MVA, 132/33 kV oil-immersed power transformer serving as the primary grid infeed for Tirunelveli industrial zone. The unit has been in continuous service since commissioning in March 2012 and has accumulated 14 years of service. The last major internal inspection (DGA and oil processing) was conducted in September 2022. Cooling fans CF-TX7-01 through CF-TX7-04 are installed for ONAF operation. The on-load tap changer (OLTC) is a Maschinenfabrik Reinhausen (MR) VACUTAP VV model with 17 tap positions, last maintained in April 2025.

2. Oil Analysis (Dissolved Gas Analysis — DGA)

Gas	Measured (ppm)	IEEE C57.104 Limit	Status
Hydrogen (H2)	85	< 100	Normal
Methane (CH4)	52	< 120	Normal
Ethane (C2H6)	41	< 65	Normal
Ethylene (C2H4)	31	< 50	Normal
Acetylene (C2H2)	8	< 3 (alarm)	Above Alarm
Carbon Monoxide (CO)	320	< 350	Normal
Carbon Dioxide (CO2)	2,840	< 2,500 (caution)	Above Caution
TDCG (Total Dissolved Combustible Gases)	597	< 720 (Level 1)	Normal

ATTENTION: Acetylene (C2H2) at 8 ppm exceeds the IEEE C57.104 alarm threshold of 3 ppm. Acetylene generation indicates arcing or high-temperature hot spots. This finding requires follow-up with a more frequent DGA sampling at 1-month intervals and possible internal inspection.

3. Electrical Test Results

Test	Value Measured	Acceptance Criteria	Status
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Insulation Resistance (HV-LV) 5000V	12,400 MΩ	> 1,000 MΩ	Normal
Insulation Resistance (HV-Earth)	9,800 MΩ	> 1,000 MΩ	Normal
Polarisation Index (10-min/1-min)	1.89	> 1.5	Normal
Tan Delta (Dielectric Loss)	0.0042	< 0.005 at 90°C	Normal
Winding Resistance (HV Phase A)	1.42 Ω	Within ±2% of factory	Normal
Winding Resistance (LV Phase B)	0.041 Ω	Within ±2% of factory	Normal
OLTC Contact Resistance	188 μΩ	< 200 μΩ	Normal
Turns Ratio (Tap 9, rated)	4.000:1	Expected 4.000:1	Normal
Partial Discharge (at 1.1 pu)	18 pC	< 100 pC	Normal
Oil BDV (Breakdown Voltage)	38 kV	> 30 kV (IEC 60156)	Normal
Oil Water Content	18 ppm	< 25 ppm	Normal
Acidity of Oil	0.09 mg KOH/g	< 0.15 mg KOH/g	Normal

4. Physical Inspection

Component	Condition	Action
Conservator Tank	No sludge, silica gel breather 60% saturated	Replace silica gel cartridge
Buchholz Relay	Tested and functional; gas pocket free	None
Pressure Relief Device	Sealed and intact	None
HV Bushings (3 units)	Clean, no tracking marks, IR scan normal	None
LV Bushings (3 units)	Hairline crack on Phase-B LV bushing shed	Monitor; plan replacement
Cooling Fans (CF-TX7-01 to 04)	All fans operational; CF-TX7-03 slightly noisy	Inspect CF-TX7-03 bearing
OLTC (MR VACUTAP VV)	Oil level OK; contacts normal; selector switch smooth	None
Surge Arresters (HV side)	All 3 units healthy — IR thermography normal	None
Oil Level Indicator	Normal — 68% level	None
Control Cabinet	All relays and meters functional; insulation good	Check control cables OK
Grounding / Earthing	Earth resistance 0.8 Ω — within spec (< 1 Ω)	None

5. Summary of Issues & Recommendations

#	Issue / Action	Priority	Target Date	Owner
1	Increase DGA sampling to monthly — Acetylene monitoring (C2H2 > 3 ppm)	High	Oct-2026	K. Anburaj
2	Replace silica gel cartridge on conservator breather	Medium	25-Sep-2026	Field Maintenance
3	Plan LV Phase-B bushing replacement at next outage	Medium	Apr-2027	Switchgear Team
4	Inspect bearing on cooling fan CF-TX7-03	Low	30-Sep-2026	Field Maintenance
5	Internal inspection (core/winding) if C2H2 continues rising	Conditional	On trigger	Chief Engineer